Athira S

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Research Interest

I am a PhD student working on the microphysics of dense matter in neutron star interiors. My research focuses on exotic degrees of freedom in neutron star cores, and their impact on the equation of state, mass–radius relations, and tidal deformability within relativistic mean-field frameworks. I have completed projects on (i) kaon condensation incorporating SU(3) flavor symmetry, (ii)r-mode instability in compact stars and its implications for the gravitational-wave event GW190814, and (iii)Bayesian analysis of hyperonic neutron-star matter within the SU(6) and SU(3) flavor-symmetry frameworks. I am currently investigating how gravitational-wave parameters can be extracted from neutron star tidal deformability, linking dense matter theory with multimessenger observations.

Education

Indian Institute of Technology Jodhpur, Rajasthan, India
Ph.D. in Physics *2023–Present*
CGPA (Coursework): 8.00
Supervisor: Dr. Monika Sinha

Sacred Heart College, Thevara, Kochi, Kerala
M.Sc. Physics *2021–2023*
CGPA: 9.08
Thesis Title: *Flares in Solar-type Stars*
Supervisor: Dr. Jeevan C. Pandey (Scientist F, ARIES)

St. Teresa's College, Kochi, Kerala
B.Sc. Physics *2018–2021*
CGPA: 9.29

Kendriya Vidyalaya Port Trust, Kochi, Kerala
Senior Secondary (Class XII, Biomaths) *2016–2018*
Board: CBSE
CGPA: 9.36

Kendriya Vidyalaya Port Trust, Kochi, Kerala
Secondary (Class X) *2016*
Board: CBSE
CGPA: 10.00

Research Experience

Department of Physics, Indian Institute of Technology Jodhpur **Rajasthan, India**
Senior Research Fellow *September 2025 – Present*

Department of Physics, Indian Institute of Technology Jodhpur **Rajasthan, India**
Junior Research Fellow *August 2023 – August 2025*

Aryabhatta Research Institute of Observational Sciences **Nainital, Uttarakhand, India**
Summer Research Fellow *June 2022 – July 2023*

Awards and Achievements

Recipient of MoE Doctoral Research Fellowship, Government of India

2023–28

M.Sc. Physics,Sacred Heart College, Thevara, Kochi, Kerala

2023

- Overall Topper
- Topper in Third Semester
- Topper in Second Semester – *Prof. G. Ramanathan Award*
- Topper in First Semester – *Prof. A. Varadarajan Award*

Recipient of Summer Research Fellowship Programme (SRFP)

2022

SRFP jointly conducted by the Indian Academy of Sciences (IASc), Bengaluru; the Indian National Science Academy (INSA), New Delhi; and the National Academy of Sciences (NASI), Allahabad.

Skills

- Languages: English, Malayalam, Hindi
- Programming: Python, Fortran
- Scientific Tools: QtGrace, Gnuplot, Excel
- Documentation: LaTeX, Microsoft Word

Publications

1.Athira S., Monika Sinha, Vivek Baruah Thapa, and Vishal Parmar. Kaon-meson coupling from su(3) flavour symmetry and application to antikaon condensed dense matter in neutron star. *Journal of Cosmology and Astroparticle Physics*, 2025(10):037, October 2025.

2.Athira S. and Monika Sinha. r-mode stabilization in rotating hyperon-rich neutron stars and its implications for gw190814. *Phys. Rev. D*, 113:063011, Mar 2026.

3.Athira S., Vishal Parmar, Monika Sinha, and Ignazio Bombaci. Neutron star matter with hyperons: Bayesian comparison of nucleonic and su(6)/su(3) hyperonic models. *Phys. Rev. C*, 114:035802, Sep 2026.

4.Athira S., Vivek Baruah Thapa, and Monika Sinha. Influence of δ resonances in hyperonic neutron stars on gravitational-wave inspiral signatures, 2026.

Presented

1. Flash talk entitled "Antikaon condensed dense matter in neutron star with SU(3) flavour symmetry" at the **GWsNS-2026: Gravitational Waves from Neutron Stars** Thematic School. Station

Biologique de Roscoff, Roscoff, France (2026).

2. Oral presentation entitled "Kaon-meson coupling from SU(3) flavour symmetry and application to antikaon condensed dense matter in neutron star" at **1st International Conference on Compact Objects Astrophysics and Astronomy**. Indian Institute of Technology, Jodhpur, Rajasthan, India (2025).
3. Poster presentation entitled "Kaon-meson coupling from SU(3) flavour symmetry and application to antikaon condensed dense matter in neutron star" at **43rd Meeting of the Astronomical Society of India**. National Institute of Technology, Rourkela, Orissa, India (2025).

References

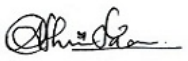
Dr. Monika Sinha: Associate Professor, Department of Physics, IIT Jodhpur, Rajasthan
Email: ms@iitj.ac.in

Dr. Jeevan C. Pandey: Scientist F, ARIES, Nainital, Uttarakhand
Email: jeevan@aries.res.in

Declaration

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

12 November 2026
Rajasthan, India



Athira S